Governance Signals for Cooperation Stability Under Adversarial Strategy Injection in Sequential Fishery Commons

Ameer Firas

March 2, 2026

Abstract

We study cooperation robustness in a minimal sequential fishery commons under adversarial strategy injection. Rather than asking whether cooperation emerges in a fixed population, we evaluate whether governance signals preserve cooperation under repeated strategy turnover and held-out regime shifts. The paper_v1 cycle uses a matched matrix across two regime tiers (easy_v1, medium_v1), two injector types (mutation and live llm_json via Ollama), and three governance conditions (none, monitoring, monitoring+sanctions). In medium_v1 high-power reruns, monitoring+sanctions reduces held-out collapse relative to none for both injectors with positive confidence bounds, supporting invasion-stability gains from minimal governance. Hard-tier results are treated as appendix-only stress evidence due to saturation.

1 Introduction

Classic commons theory predicts over-extraction under open access and highlights institutional design as the key lever for sustainable outcomes [2, 6, 10]. In machine learning, sequential social dilemma research and benchmark design emphasize policy-level dynamics, long horizons, and held-out evaluation rather than one-shot game payoffs [1, 3, 4]. Recent LLM-agent commons work has shown that cooperation can emerge in simulated societies, but robustness under distribution shift and repeated adversarial pressure remains weakly characterized [7, 8].

This work studies governance robustness in a minimal sequential fishery commons under adversarial strategy injection. The core research question is: *Which governance signals stabilize cooperation under adversarial strategy injection, and does this hold across regime difficulty and injector type (mutation versus live LLM generation)?*

The key gap is a missing *invasion-stability protocol* for commons governance. Prior commons simulations typically evaluate emergence in mostly fixed populations, while continuous adversarial generation pressure has been explored mainly in other domains (for example, program-level Red Queen dynamics) [9]. Broader policy discussions around generative-agent societies also stress realism and evaluation discipline but do not provide this protocol for renewable commons governance [5]. This leaves an open problem: measuring whether governance signals remain effective when new exploitative strategies are repeatedly generated and injected.

The evaluation is designed as a matched comparison rather than a one-off demonstration. We use the same experiment budget and protocol across governance conditions, perform selection in train regimes, and evaluate robustness in held-out regimes. Outcomes are reported with uncertainty intervals from run-level summaries.

The cycle contribution is an evidence package for invasion-stability rather than a new environment feature. Concretely, we provide a matched governance matrix over two benchmark tiers (easy_v1

and medium_v1) and two injector types (mutation and live llm_json), confidence-interval reporting for collapse reduction and supporting robustness metrics, a composite governance ranking that aggregates collapse, stock, and survival effects, and explicit gate logic that determines whether claims are supported or invalidated by regime saturation.

Hard-tier results are treated as appendix stress evidence only. Primary claims are based on medium-tier matched runs, where governance separation is measurable and non-saturated.

2 Methods

The substrate is a minimal renewable fishery commons with a shared stock, per-agent harvest actions, regeneration, and collapse dynamics. Governance mechanisms modify strategic pressure through monitoring and sanctions rather than only post-hoc accounting, so institutional choices can causally affect ecological outcomes.

Each run is evaluated under three governance settings: none, monitoring, and monitoring with sanctions. Primary contrasts are reported as $\Delta(\texttt{monitoring_sanctions} - \texttt{none})$. For collapse readability, we also report `none - monitoring_sanctions` as "collapse reduction."

Two injector sources are matched. The first is **mutation**, a non-LLM strategy variation baseline. The second is **llm_json**, where strategies are generated online by Ollama (`qwen2.5:3b-instruct`) as structured JSON and then adapted into executable policy specifications.

The core matrix spans tiers (`easy_v1`, `medium_v1`) and injectors (mutation, llm_json) with shared settings. Run counts follow the curated cycle outputs, where easy-tier cells have $n_pairs = 3$ and medium-tier high-power cells have $n_pairs = 5$.

Table 1: Shared run settings for matched comparisons.

Setting	Value
Generations	15
Seeds per generation (train)	32
Seeds per generation (test)	32
Replacement fraction	0.2
Adversarial pressure	0.3
Train regeneration rate	2.0
Train observation noise std	6.0

Selection and replacement are performed in train conditions, while governance robustness is evaluated on held-out regime packs each generation. This separates adaptation pressure from generalization assessment.

The primary outcome is held-out collapse reduction. Secondary outcomes are held-out mean stock, time-to-collapse, and per-regime survival fraction over generations. A composite governance score averages paired effect sizes (d_z) for collapse reduction, stock gain, and survival gain.

Analysis is run-level and CI reporting uses bootstrap mean confidence intervals (B=5000, percentile 2.5/97.5) over paired governance deltas. Gate logic is evaluated from frozen summary artifacts to prevent post-hoc threshold drift.

Methods and run metadata are frozen in curated outputs. The canonical files are listed below.

Table 2: Canonical curated reporting artifacts.

Artifact	Path
Methods summary	`results/runs/showcase/curated/paper_v1_methods.md`
Gate outcomes	`results/runs/showcase/curated/paper_v1_gates.json`
Main delta table	`results/runs/ablation/curated/paper_v1_table1_main_deltas.csv`

3 Results

Under matched settings with high-power reruns ($n_pairs = 5$), monitoring+sanctions reduced held-out collapse relative to none for both injector types. For mutation, collapse reduction was 0.109896 (95% CI $[0.031042, 0.189583]$). For live llm_json injection, collapse reduction was 0.083542 (95% CI $[0.015417, 0.150938]$). Both confidence-interval lower bounds are positive, supporting the medium-tier robustness claim for governance under invasion pressure.

Monitoring+sanctions also improved supporting robustness indicators versus none. Mean held-out stock increased by 23.352178 for mutation (95% CI $[15.873344, 31.697602]$) and by 22.402775 for llm_json (95% CI $[18.453585, 27.746404]$). Per-regime survival over generations improved by 0.109896 for mutation (95% CI $[0.031042, 0.189583]$) and by 0.083542 for llm_json (95% CI $[0.015417, 0.151667]$). Time to collapse increased by 1.2 generations for mutation (95% CI $[0.4, 2.2]$) and by 1.8 generations for llm_json (95% CI $[1.0, 3.0]$).

From the frozen gate file, the cycle-level criteria evaluate as follows: the primary medium reduction gate is **PASS**, the strong CI lower-bound gate is **PASS**, and the medium saturation check (> 0.95 collapse for all conditions) is **FALSE**.

Easy-tier runs used $n_pairs = 3$ and show wider uncertainty, especially for llm_json. For collapse reduction, mutation remains positive (0.170139, 95% CI $[0.068750, 0.279167]$), while llm_json crosses zero (0.163542, 95% CI $[-0.061458, 0.411979]$). This is consistent with treating medium-tier high-power evidence as the decision-critical result.

Table 3: Collapse reduction (none minus monitoring+sanctions) by tier and injector. Positive values indicate governance reduces collapse.

Tier	Injector	n_pairs	Mean reduction	95% CI
easy_v1	mutation	3	0.170139	[0.068750, 0.279167]
easy_v1	llm_json	3	0.163542	[-0.061458, 0.411979]
medium_v1	mutation	5	0.109896	[0.031042, 0.189583]
medium_v1	llm_json	5	0.083542	[0.015417, 0.150938]

Using the composite effect-size score across collapse reduction, stock gain, and survival gain, llm_json obtains a score of 1.805738 (rank-within-tier 1) and mutation obtains a score of 1.490345 (rank-within-tier 2).

Figures 1–4 visualize governance effects, time-to-collapse improvements, regime-level survival dynamics, and matched injector comparisons.

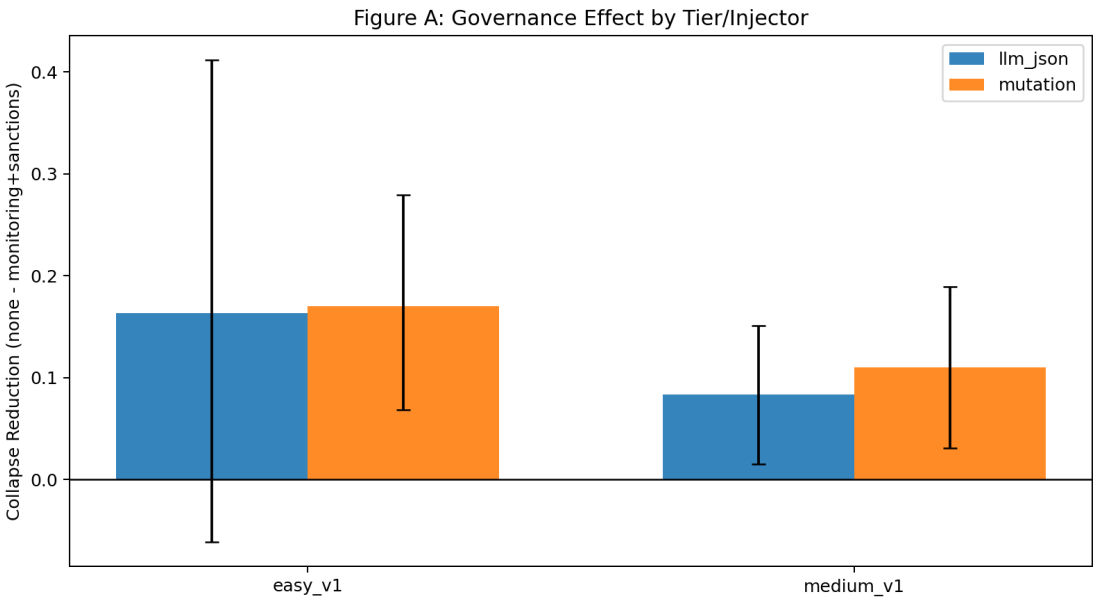

Figure 1: Governance effect by tier/injector with confidence intervals.

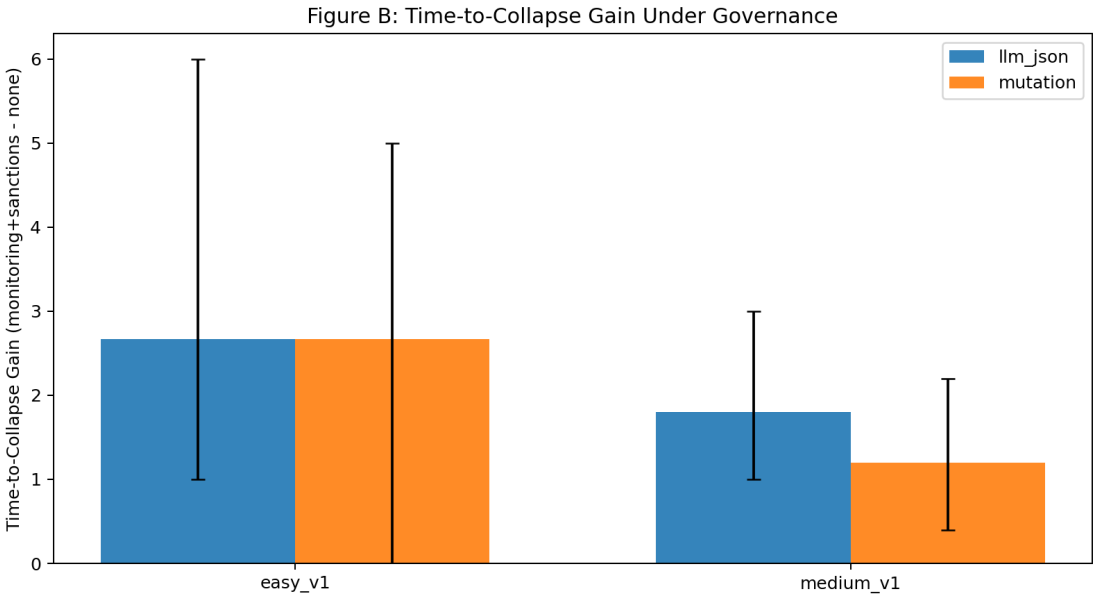

Figure 2: Time-to-collapse gain under governance.

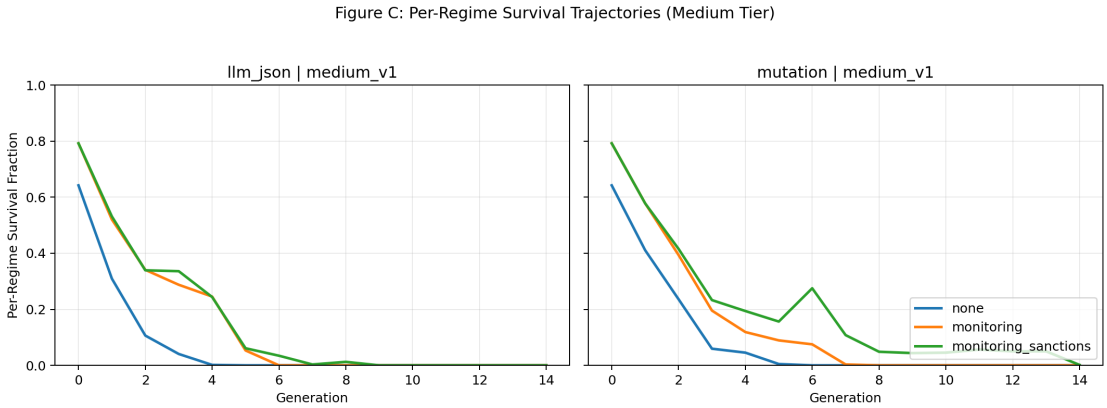

Figure 3: Per-regime survival trajectories in medium tier.

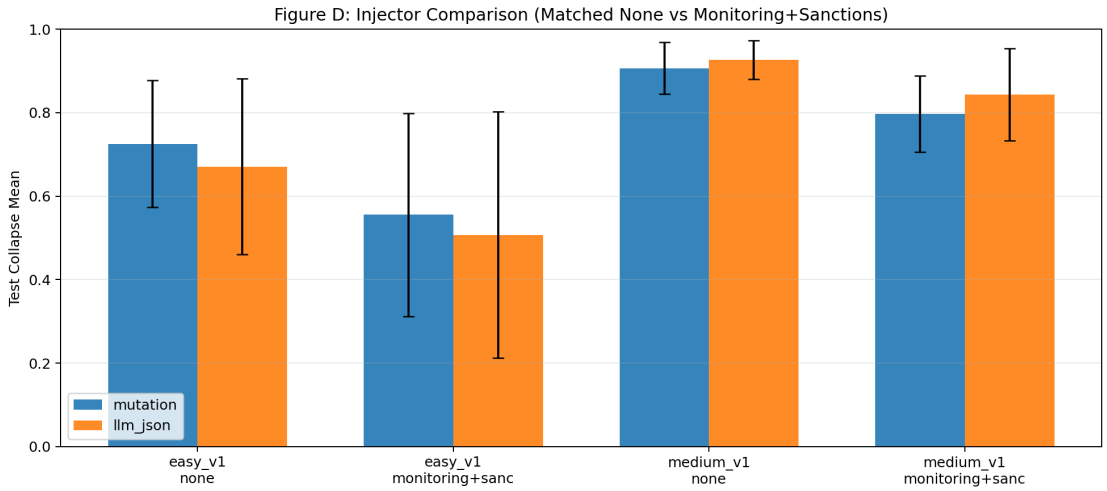

Figure 4: Injector comparison under matched settings.

4 Limitations

LLM runs are slower and can be sensitive to local runtime load. Confidence intervals are finite-sample and remain wider on easy tier. Hard_v1 shows saturation in smoke profile and is not used as primary mechanism-comparison evidence.

5 Appendix

Hard-tier stress evidence is included as appendix-only context. The appendix note source is `results/runs/showcase/curated/paper_v1_appendix_hard_note.md`.

References

[1] Yali Du et al. A review of cooperation in multi-agent learning. *arXiv preprint arXiv:2312.05162*, 2023. URL https://arxiv.org/abs/2312.05162.

[2] H. Scott Gordon. The economic theory of a common-property resource: The fishery. *Journal of Political Economy*, 62(2):124–142, 1954.

[3] Joel Z. Leibo, Vinicius Zambaldi, Marc Lanctot, Janusz Marecki, and Thore Graepel. Multi-agent reinforcement learning in sequential social dilemmas. *arXiv preprint arXiv:1702.03037*, 2017. URL https://arxiv.org/abs/1702.03037.

[4] Joel Z. Leibo et al. Scalable evaluation of multi-agent reinforcement learning with melting pot. *arXiv preprint arXiv:2107.06857*, 2021. URL https://arxiv.org/abs/2107.06857.

[5] Sarah Mercer. Welcome to willowbrook: The simulated society built by generative agents. CETaS publication, 2023. URL https://cetas.turing.ac.uk/publications/welcome-willowbrook. Published: December 16, 2023; Accessed: 2026-03-02.

[6] Elinor Ostrom. *Governing the Commons: The Evolution of Institutions for Collective Action*. Cambridge University Press, 1990.

[7] Giorgio Piatti and collaborators. Govsim: Governance of commons simulations. GitHub repository, 2024. URL https://github.com/giorgiopiatti/GovSim. Accessed: 2026-03-02.

[8] Giorgio Piatti, Zhijing Jin, Max Kleiman-Weiner, Bernhard Schoelkopf, Mrinmaya Sachan, and Rada Mihalcea. Emergence of sustainable cooperation in a society of llm agents. *arXiv preprint arXiv:2404.16698*, 2024. URL https://arxiv.org/abs/2404.16698.

[9] Sakana AI. Digital red queen: Adversarial program evolution in core war with llms. Project page, 2026. URL https://sakana.ai/drq/. Published: January 8, 2026; Accessed: 2026-03-02.

[10] Anthony Scott. The fishery: The objectives of sole ownership. *Journal of Political Economy*, 63(2):116–124, 1955.